VITYARTHI PROJECT

BANK / ATM MANAGEMENT SYSTEM

PROJECT REPORT

Programming Language: Python

Project Type: Academic / Educational

Prepared as a documentation report for the VITYARTHI PROJECT.

1. Abstract

The VITYARTHI PROJECT is a beginner-level Python application designed to represent a basic Bank/ATM Management System. The program uses PIN authentication and provides a menu-driven interface for common banking operations such as viewing an account number, depositing money, withdrawing money, and checking the bank balance. The project demonstrates fundamental programming concepts including variables, user input, loops, conditional statements, and arithmetic operations.

2. Introduction

Banking applications require a structured way to authenticate users and provide access to account-related operations. This project models a simplified version of such an interaction for educational purposes. The program begins with PIN verification and, after successful authentication, presents banking choices to the user. The source program defines account information, an initial balance, bank details, and a three-attempt authentication mechanism.

3. Problem Statement

The project aims to develop a simple command-line banking program that can authenticate a user and allow basic account operations through a menu. The application should provide a clear sequence of interaction: authenticate the user, present available operations, process the selected operation, and display the relevant result.

4. Objectives

- To develop a simple banking application using Python.
- To implement basic PIN-based user authentication.
- To limit authentication attempts to three tries.
- To provide options for viewing an account number and bank balance.
- To support deposit and withdrawal operations.
- To practice loops, conditions, variables, input, output, and arithmetic operations.
- To understand the structure of a menu-driven program.

5. Scope of the Project

The scope of this project is limited to a basic educational simulation of an ATM/banking interaction. It focuses on account authentication and simple transaction-related operations through the terminal. It does not implement a database, online banking connectivity, encrypted authentication, multiple user accounts, or real financial transactions.

6. Technologies and Concepts Used

Technology / Concept	Purpose
Python	Programming language used to implement the application.
Variables	Store PIN, account number, balance, bank name, and attempt count.

Technology / Concept	Purpose
input()	Accept information from the user.
print()	Display messages and transaction results.
while loop	Control PIN attempts and the banking menu.
if / elif / else	Check PIN and process menu choices.
Arithmetic operators	Update the account balance for deposits and withdrawals.

7. Source Code Implementation

The following screenshot shows the project implementation in the Python editor. It includes the account setup, three-attempt PIN authentication, and the beginning of the menu-driven banking operations.

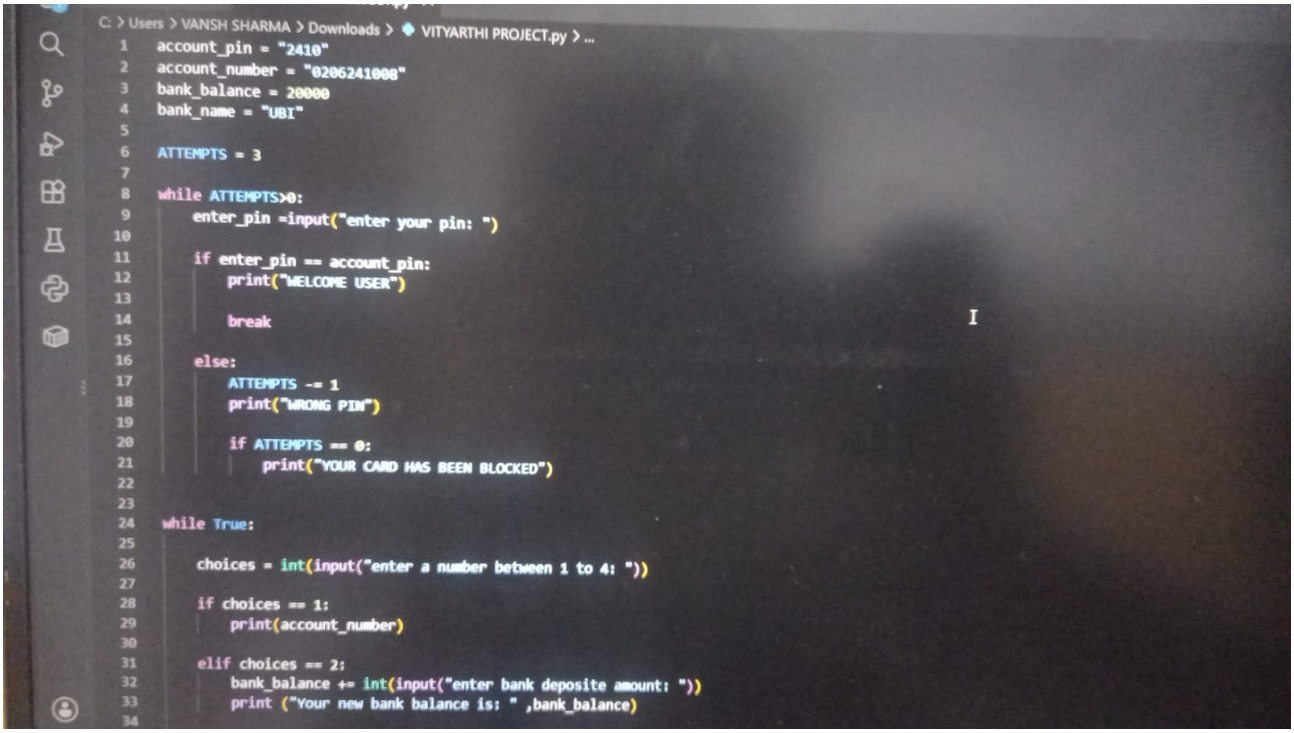


Figure 1: VITYARTHI PROJECT Python source code implementation.

8. System Design and Working

8.1 Initial Account Setup

The program initializes an account PIN, account number, bank balance, bank name, and the number of authentication attempts. These values form the initial state of the application.

8.2 Authentication Module

The authentication loop repeatedly asks the user for a PIN while attempts remain. A correct PIN results in a welcome message and exits the authentication loop. For an incorrect PIN, the attempt counter is reduced. When no attempts remain, the program reports that the card has been blocked.

8.3 Menu Module

After successful authentication, the program enters a continuous menu loop. The user enters a number from 1 to 4, and conditional statements determine which banking operation is performed.

8.4 Transaction Operations

The account-number option displays the account number. The deposit option adds the entered amount to the current balance. The withdrawal option processes an entered amount according to the implemented program logic. The balance option displays the current balance.

9. Program Flow

Flow representation:

```
START
|
v
Enter PIN
|
+-- Correct --> WELCOME USER --> Banking Menu
|
+-- Wrong --> Reduce Attempts
|
+-- Attempts remain --> Enter PIN again
|
+-- No attempts --> CARD BLOCKED

Banking Menu
|
+-- 1 --> Display Account Number
+-- 2 --> Deposit Amount
+-- 3 --> Withdrawal
+-- 4 --> Display Balance
+-- Other --> Wrong Input Message
```

10. Sample User Interaction

A typical successful interaction begins with the user entering the configured PIN. The program then displays the banking menu. For example, selecting the balance option displays the current balance.

```
enter your pin: 2410

WELCOME USER

enter a number between 1 to 4: 4

20000
```

11. Expected Outcomes

- The program accepts a PIN from the user.
- A correct PIN grants access to the banking menu.
- Incorrect PIN entries reduce the remaining attempt count.
- After three failed authentication attempts, the program reports that the card is blocked.
- The account number can be displayed.
- Deposits update the stored balance.
- The withdrawal operation processes the entered amount according to the program logic.
- The current balance can be displayed.

12. Testing Overview

Test Case	Input / Condition	Expected Result
Correct PIN	Correct configured PIN	WELCOME USER and access to menu
Wrong PIN	Incorrect PIN	WRONG PIN and attempts decrease
Three wrong attempts	Incorrect PIN three times	Card blocked message
Account number	Menu option 1	Account number displayed
Deposit	Menu option 2 + amount	Balance is increased

Test Case	Input / Condition	Expected Result
Withdrawal	Menu option 3 + amount	Withdrawal logic is executed
Balance	Menu option 4	Current balance displayed
Invalid menu input	Number other than 1–4	Wrong-input message

13. Limitations

- Account credentials and account information are stored directly in the source code.
- The application is command-line based and has no graphical user interface.
- There is no persistent database for storing account transactions.
- The project is not connected to a real banking system.
- The project contains simplified transaction logic and is intended for learning rather than financial use.

14. Security Considerations

Because this is an educational program, sensitive information is directly represented in the source code. A production banking application would require secure credential handling, PIN protection, encryption, authorization controls, transaction validation, database security, logging, and other safeguards. The current project should therefore be treated only as a learning demonstration.

15. Learning Outcomes

By completing this project, a learner can understand how to combine basic Python constructs into a small working application. The project provides practical experience with input/output, variables, loops, conditions, arithmetic operations, and menu-driven program design.

16. Conclusion

The VITYARTHI PROJECT demonstrates the basic structure of an ATM-style banking application using Python. It combines authentication, menu selection, and simple account operations in a single program. Although the system is intentionally simple and not suitable for real banking, it provides a useful foundation for learning programming logic and for developing more advanced applications with databases, graphical interfaces, and stronger security.

17. Project Details

Project Name	VITYARTHI PROJECT
Application	Bank / ATM Management System
Programming Language	Python
Project Level	Beginner / Academic
Interface	Command Line
Purpose	Educational demonstration of basic programming concepts